Elements of Soliton Theory; Solitons: Mathematical Methods for Physicists; and Solitons and the Inverse Scattering Transform

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Citation: Physics Today **35**, 6, 55 (1982); doi: 10.1063/1.2915133
View online: http://dx.doi.org/10.1063/1.2915133
View Table of Contents: http://physicstoday.scitation.org/toc/pto/35/6
Published by the American Institute of Physics



Among the first texts to explain the "soliton revolution"

Elements of Soliton Theory
G. L. Lamb Jr
301 pp. Wiley, New York, 1980. $29.95

Solitons: Mathematical Methods for Physicists
G. Eilenberger
192 pp. Springer, New York, 1981. $29.00

Solitons and the Inverse Scattering Transform
M. Ablowitz, H. Segur
425 pp. SIAM, Philadelphia, 1981. $54.50

Reviewed by Chia Hsiung Tze



Oblique interaction of two shallow water waves, with a resulting phase shift of each wave crest. The two-soliton solution of the Kadomtsev–Petviashvili equation is qualitatively similar to this interaction. The photograph, taken by Terry Toedtemeier off the coast of Oregon near Manzanita, appears in *Solitons and the Inverse Scattering Transform* reviewed here.

The last decade has witnessed a revolution in the understanding of nonlinear phenomena in many branches of science. Most surprising has been the discovery of a single methodology—conceptual and mathematical—that can solve a great diversity of nonlinear problems. Whether we survey condensed-matter physics, statistical physics, or classical and quantum field theories of matter and gravity, we cannot fail to notice the concept of a soliton (a nondispersing wave packet with particle-like properties), elevated by some to a paradigm in this era of nonlinear sciences.

While J. Scott Russell first observed solitons in water waves in August 1834, a full-fledged theory of solitons has only come of age in the last decade. This advance is due primarily to the discovery of a generalization of the Fourier transform, the inverse scattering transform. While this method can be used to solve exactly only a certain number of nonlinear equations, many of these are relevant to broad areas in physics. In contrast to the inadequacy of traditional methods of mathematical physics in tackling the intrinsically nonperturbative effects of nonlinear systems, the inverse scattering transform has supplied material for a new and powerful arsenal of mathematical tools forged by mathematicians since the last century. The mathematics necessary for the transform includes the spectral theory of pseudodifferential operators, global differential and algebraic geometries and the representation theory of infinite-dimensional affine Lie algebras. Indeed many of the actual computations of integrable equations involve significant amounts of formal mathematics because these solvable equations are seen as zero-order approximations to real systems—counterparts to the harmonic oscillator in linear systems.

Thus there is an urgent need to convey the substance and the special mathematical techniques as well the scope and excitement of this "soliton revolution" to new generations of scientists. Until recently a newcomer to soliton theory had to learn directly from the numerous published papers and conference proceedings or from the few comprehensive review articles devoted to this topic. At long last a few leading practitioners in the field here and abroad have taken the time out to write the first texts on solitons: The three volumes under review are part of this timely endeavor. Obviously this brief review can in no way do full justice to even one—much less three—books on so broad a topic as the theory of solitons.

In the past decade, George L. Lamb Jr has contributed importantly to the understanding of both the physics and the aesthetics underlying geometry and Backlund structure of soliton equations. His *Elements of Solitons* provides a masterfully balanced coverage of a set of basic nonlinear phenomena and the minimal mathematical formalism needed for their description. Chapter 1 goes right to the very heart of the subject: It establishes the key connection between a canonical nonlinear evolution equation, the Korteweg–de Vries equation, and a linear Schrödinger equation. Indeed the existence of a mapping to a linear problem is the universal trait shared by a large class of nonlinear systems. This simple, rather miraculous structural feature allows for their exact integrability and constitutes the major theme of all three volumes under review. Chapters 2 and 3 of Lamb's book treat one-dimensional direct and inverse scattering in a thorough and flowing manner that is typical of the whole book. Chapter 4 illustrates the formalism with both numerical and analytic discussions of multisoliton solutions along with a much too cursory treatment of the infinite number of local conservation laws for the Kortweg–de Vries system. Chapter 5 follows with a compact introduction to other standard soliton equations, such as the "sine–Gordon" and nonlinear Schrödinger equations. The following two chapters put all these

equations in selected physical contexts that range from water and plasma waves for the Kortweg–de Vries and nonlinear Schrödinger equations to dislocation theory for the sine–Gordon equation. Chapter 8 introduces the important and unavoidable topic of Backlund transformations, the nonlinear counterpart of the superposition rule for solutions in linear systems. Finally, Chapter 9 covers soliton perturbation theory, which often serves as the connecting link between the exactly solvable theories and real physical systems. Each topic is lucidly presented by reference to a readily understandable specific physical example. This approach eases the way for the stepwise introduction of relevant mathematical formalisms. Supplementing the textual materials are a few relevant exercises offered at the end of most sections. Advanced undergraduates and entering graduate students of physics, applied sciences and mathematics alike will enjoy learning from this fine, self-contained primer. I have only one *caveat*: Watch out for over a dozen misprints that occur mostly in the formulae and their numbering!

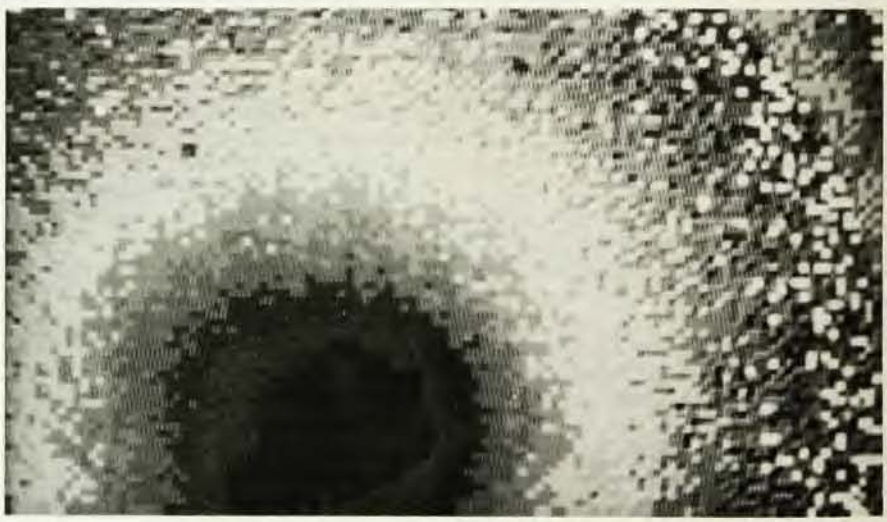
Solitons: Mathematical Methods for Physicists is a well-written, compact monograph, a fine translation from the original German. It derives from a graduate-level theoretical physics course taught by its author, Gert Eilenberger, at the University of Cologne. While the book rightly assumes some familiarity with differential equations and function and operator theories, a conscientious reader, with pencil and paper in hand, will benefit most from this highly condensed treatment. Definitely more formal in content and set at a somewhat higher level than Lamb's book, this volume does succeed in giving a self-contained introduction to the mathematics of solitons in one space dimension.

After concisely stating the necessary physical motivations, the book focuses on the methods of inverse scattering. It treats the Kortweg–de Vries system first as a Lagrangian field theory with solitons and higher conservation laws, then as a detailed illustration of the inverse scattering transform, which is further developed with other soliton equations. A chapter is devoted to the sine–Gordon system, the canonical example of a nontrivial integrable relativistic field theory in two-dimensional space–time. There follows an introduction to the statistical mechanics of the sine–Gordon system, a topic of much current experimental import in condensed-matter physics. Finally, for good contrast, Eilenberger further illustrates the inverse scattering transform for a discrete system, the Toda lattice, an object of fascination for both mathematicians and physicists. He gives a selected list of annotated references that will facilitate further study. Eilenberger has done an excellent job of summarizing the essential aspects of current soliton theory and has succeeded in conveying the structural beauty and scope of the subject. Graduate students in physics willing to work diligently through the book will profit much from this solid introduction to a topical area of research. It would make an excellent text for a short concentrated course or a supplementary reading for a longer one.

Solitons and the Inverse Scattering Transform is written by Mark J. Ablowitz and Harvey Segur, among the foremost authorities in the field. They intend their book to be an up-to-date compendium and a comprehensive guide to the many theoretical and mathematical ideas about solitons developed over the last decade. The book focuses on the mathematical intricacies of the inverse scattering transform and, to a lesser extent, on related approaches to solving integrable equations. It begins with a concise, highly informative, historical introduction to solitons. It then presents in great detail the formalism of the inverse scattering transform, upon which the materials of subsequent chapters are built. Chapter 2 covers the transform in other settings: higher-order eigenvalue problems and scattering problems in higher dimensions, the Toda lattice, and finite-band solutions (specifically of the Kortweg–de Vries equation). Chapter 3 surveys methods other than the inverse-scattering transform, covering topics such as Backlund transformations, Hirota's method for N-soliton solutions, and Painlevé transcendents as well as soliton perturbation theory. Chapter 4 offers selected physical applications of the transform—discussed at varying lengths—to hydrodynamics, nonlinear optics, axisymmetric gravitation and quantum field theory. Each chapter closes with a rich fund of exercises, some of which are actually challenging research problems. Also a helpful "Appendix of the Fourier Transform" is provided. This impressive book—with its exhaustive bibliography of over 500 articles—will serve equally well as an advanced text and an authoritative reference guide to the mathematics of solitons. It is ideally suited for engineers, physicists and applied mathematicians who desire to acquire a broad but quite detailed view of this fascinating and explosive area of research.

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Chia-Hsiung Tze, a member of the physics department of Yale University, is interested in the mathematical theories of solitons.



The star Sirius in an x-ray image produced by the fine-error sensor of the International Ultraviolet Explorer satellite on 20 September 1978. (Photograph taken by Fred Espenak.)

X-ray Astronomy

J. L. Culhane, P. W. Stanford
Scribner's, New York, 1981. $25.00

The study of x-ray emissions by nonterrestrial sources has undergone dramatic advances since it began in the late 1940s with the positive detection of solar x-radiation. During the past three decades instrumentation has progressed from simple photographic detectors and Geiger counters briefly sent aloft on sounding rockets to a sophisticated focusing x-ray telescope equipped with two-dimensional imaging detectors aboard the HEAO-2 ("Einstein") satellite. A wide variety of astronomical objects are now known to produce x rays, including ordinary main-sequence stars, white dwarfs and neutron stars, active galaxies, clusters of galaxies, and quasars. X-ray astronomy has also provided the first tentative identification of a black hole in the x-ray